

# Methodology for Deriving Lunar Elemental Ratios from Chandrayaan-2 CLASS X-ray Spectra

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**Abstract**—This document proposes a methodology for deriving lunar elemental abundance ratios from Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) data. The pipeline accounts for particle background, solar illumination, scattered solar continuum, instrumental response, X-ray fluorescence (XRF) production, and compositional matrix effects before estimating elemental abundances. Mg/Si and Al/Si are expected to be the most robust ratios, while Ca/Si, Fe/Si, and Ti/Si are treated conditionally. Statistical and systematic uncertainties and external validation are also considered.

## I. INTRODUCTION

The Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) measures lunar X-ray fluorescence (XRF) produced when solar X-rays illuminate the lunar surface. Characteristic fluorescence lines can be used to infer the abundances of surface elements such as Mg, Al, Si, Ca and Fe.

The important point is that the measured intensity of an XRF line is not determined solely by the abundance of the corresponding element. It also depends on the incident solar spectrum, geometry, compositional matrix effects, and instrumental response. Therefore, reliable elemental ratios cannot be obtained by directly taking ratios of peak heights or raw peak areas.

This document therefore proposes a physics-based spectral-fitting pipeline that converts the observed CLASS spectrum into elemental abundances and subsequently into abundance ratios.

## II. OVERVIEW OF THE PROPOSED PIPELINE

The complete pipeline is:

- 1) Read and calibrate the CLASS L1 spectrum.
- 2) Verify the observation time, footprint and observing geometry.
- 3) Combine appropriate time bins to obtain sufficient statistics.
- 4) Screen the observation for particle contamination and anomalous background.
- 5) Obtain the corresponding solar spectrum from XSM.
- 6) Obtain a suitable lunar-orbit background spectrum.
- 7) Subtract or model the instrumental/particle background.
- 8) Construct the solar-scattered continuum model.
- 9) Identify the relevant XRF lines.
- 10) Construct a physics-based XRF forward model.
- 11) Fold the model through the CLASS RMF and ARF.
- 12) Fit the observed spectrum simultaneously with the continuum and XRF components.

- 13) Estimate elemental abundances and their covariance.
- 14) Calculate elemental ratios such as Mg/Si and Al/Si.
- 15) Propagate statistical and systematic uncertainties.
- 16) Validate the results using external datasets and alternative solar-spectrum models.

## III. RAW CLASS DATA PREPARATION

### A. Time and Footprint Verification

Each CLASS spectrum must first be verified against the specified UTC interval, footprint and observing geometry. This is important because the incident solar spectrum varies with time, particularly during solar flares. Spectra should therefore be selected using their observation timestamps rather than filenames alone.

### B. Energy Calibration

The CLASS channel coordinate is converted to photon energy using the instrument calibration. This is essential because elemental identification is based on characteristic X-ray energies. The principal XRF lines of interest are Mg-K $\alpha$ , Al-K $\alpha$ , and Si-K $\alpha$ , with higher-energy lines such as Ca-K $\alpha$ , Ti-K $\alpha$ , and Fe-K $\alpha$  considered when sufficiently excited and detected.

### C. Time Binning

Individual 8-second spectra may have insufficient photon statistics for reliable line fitting, so adjacent spectra may be integrated. The integration interval should balance improved photon statistics against temporal changes in the incident solar spectrum. The reference CLASS analysis used 96-second integrations for abundance estimation.

## IV. BACKGROUND AND DATA QUALITY TREATMENT

### A. Particle Background

CLASS detects lunar XRF together with particle-induced instrumental background, including contributions from Galactic Cosmic Rays. A nearby night-side observation from the same orbit can be used to characterize the particle background and should be scaled appropriately to the science exposure before subtraction or simultaneous modelling.

### B. Particle Contamination Screening

Background subtraction alone is insufficient because the particle background can vary with the spacecraft environment, particularly during geotail passages. The 7–16 keV count rate should therefore be monitored as a diagnostic of changes in particle flux.

Intervals showing anomalous particle background should be rejected or treated separately. Persistent instrumental features, particularly Al-K $\alpha$  from the optical blocking filter and Cu-K $\alpha$  from the collimator, must also be distinguished from genuine lunar fluorescence.

## V. SOLAR SPECTRUM DETERMINATION

### A. Importance of the Incident Solar Spectrum

The incident solar spectrum determines the photons available to excite each element and therefore strongly affects XRF line strengths. This is particularly important for higher-energy lines such as Ca-K $\alpha$ , Ti-K $\alpha$ , and Fe-K $\alpha$ .

### B. XSM Spectrum

The preferred solar input is the simultaneous Solar X-ray Monitor (XSM) spectrum corresponding to the CLASS observation. XSM provides the solar spectrum at high spectral and temporal resolution, allowing changes in the solar flare spectrum to be incorporated into the XRF calculation. The XSM spectrum should therefore be matched to each CLASS integration interval rather than assuming constant solar illumination.

## VI. CONSTRUCTION OF THE SPECTRAL MODEL

After background treatment, the lunar spectrum should not be modelled as a collection of isolated Gaussian peaks alone.

The observed spectrum can conceptually be represented as

$$S_{\text{obs}}(E) = S_{\text{background}}(E) + S_{\text{scatter}}(E) + S_{\text{XRF}}(E), \quad (1)$$

where

- $S_{\text{background}}$  is the particle/instrumental background,
- $S_{\text{scatter}}$  is the solar spectrum scattered from the lunar surface, and
- $S_{\text{XRF}}$  is the fluorescence spectrum produced by the lunar elemental composition.

### A. Scattered Solar Continuum

A fraction of the incident solar radiation is scattered by the lunar surface and contributes a continuum to the observed spectrum.

The scattered component should therefore be generated using the measured solar spectrum and the appropriate lunar scattering model rather than being represented by an arbitrary polynomial.

This is important because an incorrect continuum directly biases the estimated areas of nearby XRF lines.

### B. XRF Line Model

The characteristic XRF lines should be generated using a physics-based XRF model. The principal lines considered are:

- Mg-K $\alpha$   $\sim$  1.25 keV,
- Al-K $\alpha$   $\sim$  1.49 keV,
- Si-K $\alpha$   $\sim$  1.74 keV,
- Ca-K $\alpha$   $\sim$  3.6 keV,
- Ti-K $\alpha$   $\sim$  4.5 keV,
- Fe-K $\alpha$   $\sim$  6.4 keV.

The model calculates their expected fluorescence intensities from the assumed elemental composition and incident solar spectrum. Fe-L emission may also contribute at lower energies.

## VII. INSTRUMENT RESPONSE

The theoretical spectrum cannot be compared directly with the observed counts. The detector modifies the incident spectrum through its effective area and energy redistribution.

The model should therefore be folded through:

- 1) the Ancillary Response File (ARF), representing the effective collecting efficiency; and
- 2) the Redistribution Matrix File (RMF), representing the mapping between incident photon energy and measured detector channels.

The predicted detector spectrum can be written schematically as

$$C_i^{\text{model}} = \int R_i(E) S_{\text{model}}(E) dE, \quad (2)$$

where  $R_i(E)$  represents the instrument response.

The CLASS RMF and ARF should therefore be included in the forward-fitting process.

## VIII. FORWARD XRF MODELLING

### A. Why Peak Ratios Are Insufficient

A naive estimate such as

$$\frac{\text{Mg}}{\text{Si}} \approx \frac{I_{\text{Mg}}}{I_{\text{Si}}} \quad (3)$$

is not generally valid.

The XRF intensity of an element is affected by:

- incident solar flux,
- elemental abundance and fluorescence yield,
- absorption and enhancement by other elements,
- lunar geometry and matrix effects,
- detector response.

Consequently, the desired abundance ratio must be obtained from the composition parameters of a forward XRF model rather than from raw peak ratios.

### B. Fundamental Parameter Approach

A suitable approach is to use a fundamental-parameter XRF model.

The model takes the elemental composition and incident solar spectrum as inputs and predicts the resulting XRF line intensities and scattered continuum.

The composition is varied until the response-folded model provides the best match to the observed CLASS spectrum.

This approach is consistent with the CLASS analysis described by Narendranath et al., where an XRF model based on fundamental-parameter calculations is implemented in XSPEC/PyXSPEC.

## IX. SPECTRAL FITTING

The fitting should be performed simultaneously over the relevant energy range rather than fitting each peak independently.

A schematic fit statistic is

$$\chi^2 = \sum_i \frac{(C_i^{\text{obs}} - C_i^{\text{model}})^2}{\sigma_i^2}, \quad (4)$$

or, when count statistics are sufficiently Poisson-dominated, an appropriate Poisson likelihood such as the Cash statistic may be used.

The fit parameters should include the elemental abundances and any parameters required to describe the scattered solar continuum and instrumental/background normalization.

### A. Fit Quality

Fit quality should be assessed using the fit statistic, residuals around XRF lines and continuum regions, and stability under reasonable model changes. The reference study retained fits with reduced  $\chi^2$  between approximately 0.8 and 2, although a good global statistic is insufficient if structured residuals remain around an individual elemental line.

## X. ELEMENTAL ABUNDANCE ESTIMATION

The spectral fit provides elemental abundances such as  $A_{\text{Mg}}$ ,  $A_{\text{Al}}$ ,  $A_{\text{Si}}$  and  $A_{\text{Ca}}$ . For mass or weight abundances, an elemental ratio is given by

$$R_{A/B} = \frac{A_A}{A_B}. \quad (5)$$

Thus, the requested ratios include Mg/Si, Al/Si and Ca/Si.

The report must specify whether each ratio is a mass ratio or an atomic ratio. For atomic ratios,

$$\left( \frac{N_A}{N_B} \right) = \frac{A_A/M_A}{A_B/M_B}, \quad (6)$$

where  $M_A$  and  $M_B$  are the atomic masses.

## XI. WHICH RATIOS ARE RELIABLY DERIVABLE?

### A. Mg/Si

Mg/Si is a primary ratio because Mg- $K\alpha$  and Si- $K\alpha$  lie in the well-measured 1–2 keV region. It should nevertheless be derived from the forward model rather than direct peak-area division.

### B. Al/Si

Al/Si is similarly well suited to CLASS because Al- $K\alpha$  and Si- $K\alpha$  occur near 1.49 and 1.74 keV. It should be reported when both lines have adequate signal-to-background and the spectral fit is well constrained.

### C. Ca/Si

Ca/Si is conditional because Ca- $K\alpha$  near 3.6 keV requires sufficiently energetic incident solar photons. A non-detection therefore does not imply negligible Ca abundance. Ca/Si should only be reported when the Ca line is significantly detected and the XSM spectrum provides sufficient high-energy excitation; otherwise an upper limit or external constraint should be used.

### D. Fe/Si

Fe/Si is conditional because Fe-L is more susceptible to low-energy background and modelling degeneracies, while Fe-K requires a harder solar spectrum. It should therefore only be reported when the relevant feature is clearly detected and the fit is sensitive to Fe.

### E. Ti/Si

Ti/Si is conditional because Ti- $K\alpha$  near 4.5 keV requires sufficiently energetic solar photons. It should only be derived when Ti is statistically detected and the high-energy solar spectrum is adequately characterized.

### F. O/Si

O/Si should not be considered reliably derivable from CLASS alone under the assumptions of this analysis. The O- $K\alpha$  region is not sufficiently constrained by the available solar spectrum, so an external constraint or different treatment would be required.

## XII. RECOMMENDED RATIO CLASSIFICATION

The ratios can therefore be classified as follows:

TABLE I  
EXPECTED RELIABILITY OF ELEMENTAL RATIOS

| Ratio | Classification | Condition                                 |
|-------|----------------|-------------------------------------------|
| Mg/Si | Primary        | Adequate Mg and Si detection              |
| Al/Si | Primary        | Adequate Al and Si detection              |
| Ca/Si | Conditional    | Significant Ca detection                  |
| Fe/Si | Conditional    | Significant Fe detection                  |
| Ti/Si | Conditional    | Strong high-energy flare and Ti detection |
| O/Si  | Not CLASS-only | Requires external constraint              |

## XIII. UNCERTAINTY ESTIMATION

Uncertainty must include both statistical and systematic contributions.

### A. Statistical Uncertainty

Photon counting follows approximately Poisson statistics.

The covariance matrix returned by the spectral fit should be used to obtain the uncertainty and covariance of the fitted elemental abundances.

For a ratio

$$R = \frac{A}{B}, \quad (7)$$

the first-order propagated variance is

$$\sigma_R^2 = \left(\frac{1}{B}\right)^2 \sigma_A^2 + \left(\frac{A}{B^2}\right)^2 \sigma_B^2 - 2\frac{A}{B^3} \text{Cov}(A, B). \quad (8)$$

The covariance term should not be ignored because elemental abundances are coupled through the compositional model.

### B. Monte Carlo Propagation

A more robust approach is to perform Monte Carlo simulations.

For each realization:

- 1) perturb the observed spectrum according to its counting uncertainties;
- 2) perturb the background within its uncertainty;
- 3) perturb the solar spectrum within its measurement uncertainty;
- 4) refit the spectrum;
- 5) calculate the elemental ratios.

The resulting distribution of each ratio directly provides confidence intervals and naturally handles correlations and nonlinearities.

### C. Systematic Uncertainties

The systematic error budget should include:

- uncertainty in the solar spectrum;
- uncertainty in the background model;
- CLASS RMF/ARF calibration uncertainty;
- uncertainty in viewing and illumination geometry;
- uncertainty from the assumed compositional matrix;
- uncertainty caused by weak or undetected elements that must be externally constrained;
- model dependence of the scattered solar continuum.

## XIV. EXTERNAL VALIDATION

The derived abundances and ratios should be cross-checked against independent measurements.

### A. Lunar Prospector GRS

Lunar Prospector Gamma Ray Spectrometer (LP-GRS) elemental abundance maps provide an independent comparison. Spatial scale and sampling-depth differences must be considered, since XRF probes the upper tens of microns while GRS samples much deeper material.

### B. Chandrayaan-1 C1XS

C1XS provides an independent X-ray comparison where the CLASS footprint overlaps previously observed regions.

### C. Lunar Soil Samples

Apollo, Luna and other lunar sample measurements can provide an additional geochemical reference, although their spatial scale and sampling depth differ from orbital XRF measurements.

For each reported ratio, the statistical and systematic uncertainties, detection significance, integration interval, fit statistic, solar-spectrum source, and any external abundance constraints should be recorded. Results may be reported as

$$\text{Mg/Si} = R_{\text{Mg/Si}} \pm \sigma_{\text{stat}} \pm \sigma_{\text{sys}}. \quad (9)$$

## XV. CONCLUSION

The proposed methodology avoids interpreting raw XRF peak intensities directly as elemental abundances. The central step is a response-folded, physics-based forward model in which the measured solar spectrum, lunar scattering, elemental composition, instrumental response and background are modelled simultaneously.

Mg/Si and Al/Si are expected to provide the most robust ratios from the available CLASS spectra. Ca/Si, Fe/Si and Ti/Si should only be reported when their corresponding features are significantly detected and the incident solar spectrum provides sufficient excitation. Oxygen should not be treated as a CLASS-only abundance under the assumptions of the reference analysis.

Finally, both statistical and systematic uncertainties must be propagated, and the results should be cross-checked against independent datasets such as LP-GRS, C1XS and lunar sample measurements.